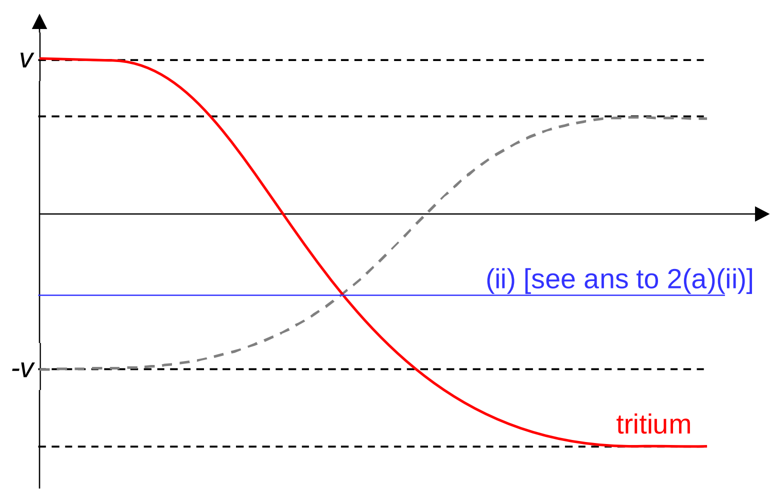
		accuracy to question)
2(a)(i)	no initial total momentum of system is not zero, when system of alpha interacting with tritium has no net external force, final total momentum of system can never be zero / centre of mass of system has a constant velocity OR by Newton's 3rd Law, magnitude of force on each particle same each particle experiences different magnitudes of deceleration and so comes to instantaneous rest at different times.	**B1** (contextualize definition of PCLM to question)

Qns	Answer	Marks
2(a)(ii)	Let m_α: mass of alpha particle m_t: mass of tritium particle v_α: initial speed of alpha particle v_t: initial speed of tritium particle v_f: final speed of combined particle $m_\alpha v_\alpha - m_t v_t = (m_\alpha + m_t) v_f$ $v_f = \frac{(4u - 3u)v}{7u} = \frac{1}{7} = 0.143v$	**A1**
2(a)(iii)	At (a)(ii), the some of the kinetic enegy has been converted to electric potential energy. As the particles repel and separate, the <u>electric potential energy will be converted back to kinetic energy.</u> Hence, the <u>kinetic energy BEFORE and AFTER the collision will be conserved</u>, and the interaction is elastic. (when electric force of repulsion is acting on both particles, the interaction/"collision" is still taking place)	**B1** Discussion on EPE converting back to KE. **B1** KE before and KE after
2(b)		**B1**

Qns	Answer	Marks
2(c)	Let $v_{f\alpha}$:final speed of alpha particle v_{ft}: final speed of tritium particle Take direction to the right as positive. For elastic collision, relative speed of approach = relative speed of separation $v_{\alpha} - v_{t} = v_{ft} - v_{f\alpha}$ $v - (-v) = v_{ft} - v_{f\alpha}$ $2v = v_{ft} - v_{f\alpha}$ -------- (1) By conservation of momentum: $m_{\alpha}v + m_{t}(-v) = m_{t}v_{ft} + m_{\alpha}(-v_{f\alpha})$ $4uv - 3uv = 3uv_{ft} - 4uv_{f\alpha}$ ----------- (2) $v = 3v_{ft} - 4v_{f\alpha}$ Using GC: $v_{f\alpha} = 0.714\ v$ $v_{ft} = 1.29\ v$	**M1** (use of property of elastic collision) **A1** (do not accept fraction) **A1** (do not accept fraction)
3(a)	Since aeroplane is constantly changing direction, and hence velocity	**B1**